UK Medicine Trade: Trend Analysis, 2022–2026

A briefing on UK medicine exports and imports, from HMRC customs data

Emanuel Cunha

November 2026

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1. ASK PHASE

Business Task

Written from the perspective of an analyst in a UK government trade department. The task: analyse the UK's customs records to show whether the UK is selling more or fewer medicines abroad than before, who is buying them, and whether it is leaning more on imports.

Problem

The Budget is on 28 October, when the government announces its tax and spending plans. Life sciences is one of its priority growth sectors, so the minister needs a clear picture of medicines trade beforehand. Monthly figures swing too much to show the trend by eye, so it takes analysis to tell whether trade is rising or falling. The data was downloaded on 9 October 2026, so in this scenario the Budget is still ahead.

Questions

The minister asks one question in three parts. Each part is measured as follows:

1. Selling more or fewer medicines abroad? How has the monthly value of exports changed since January 2022?
2. Who is buying? Which countries take the largest share of exports, and has that changed?
3. Leaning more on imports? Is the gap between exports and imports widening or narrowing, and who supplies the imports?

Metrics

The value (£) of medicines exported and imported each month, and the gap between them. Above zero, the UK sells more than it buys. Below zero, the reverse. To smooth those swings, trends are read on 12-month averages and year-on-year change. Weight (kg) is tracked alongside value, as a rough check on whether a change comes from selling more goods or from higher prices. Customers and suppliers are measured by each country's share of the total.

Scope

Trade in goods only: medicines that physically cross the UK border, as recorded by customs. "Medicines" means HS chapter 30, the international product code for pharmaceutical products. It covers medicines, vaccines and similar biological products, plus a small share (about 4% of value) of other items, such as bandages, sutures and ostomy appliances. The period is January 2022 to July 2026, month by month. Out of scope: what medicines cost the NHS, and what caused any change. Customs records show what moved, not why.

Data

HMRC's Overseas Trade in Goods Statistics: monthly UK customs records, published as official statistics under the Open Government Licence.

Stakeholders

The minister, who uses the answer to brief the Treasury before the Budget. The answer helps the minister decide what to argue for: protecting export markets, or backing UK manufacturing. The analysis informs that briefing. It does not decide the spending.

The audience is the minister and Treasury officials. They are not data specialists and have little time, so findings are presented in plain language, one chart per point, with a one-page summary.

Deliverables

A Python notebook (the full working), an interactive Power BI dashboard, a one-page executive brief, documentation and a short slide deck.

2. PREPARE PHASE

Data Sources

HMRC's Overseas Trade in Goods Statistics, the UK's official monthly record of goods trade, built from customs declarations. Two extracts were downloaded on 9 October 2026 from HMRC's Build an overseas trade data table tool. Both are filtered to HS chapter 30 (pharmaceutical products), all countries and all four trade flows (EU and non-EU, exports and imports).

The period starts in January 2022, when HMRC changed how it records EU imports. Earlier months are not comparable.

Location & Structure

Two CSV files, kept untouched in Google Drive, so the original data is always there to check back against.

File	One row per
`HMRC Medicines by **Product** (Jan 2022 - Jul 2026).csv`	• Product code • Trade flow • Month
`HMRC Medicines by **Country** (Jan 2022 - Jul 2026).csv`	• Country • Trade flow • Month

Both hold value (£), net weight (kg), trade flow, year and month.

Credibility & Licensing

Official statistics, counted almost entirely from customs declarations that traders must file by law. The data meets ROCCC: **reliable** (accredited official statistics), **original** (straight from HMRC), **comprehensive** (every month, all countries, all four flows), **current** (latest month July 2026) and **cited** (HMRC publishes a methodology report). It is published under the Open Government Licence v3.0, free to reuse with credit. No personal data is involved.

Integrity Checks

Before cleaning, three checks passed: the CSV totals match HMRC's on-screen totals, every flow has all 55 months, and the two files agree for every flow and month.

Limitations

- **Recent months can change.** HMRC revises provisional figures every month, so the files are a snapshot from 9 October 2026.
- **The gap is a guide, not an exact balance.** Imports include the cost of shipping and insurance to the UK (CIF). Exports are valued at the point they leave the UK (FOB). HMRC explains both in section 8.2 of its methodology report.
- **Other sources will not match.** ONS figures use a different basis, which HMRC explains in section 8.4 of its methodology report.
- **Import countries are countries of dispatch.** HMRC records an import under the country it was exported from to come to the UK, which may not be where it was made.

3. PROCESS PHASE

Tools

Python (pandas) in a Google Colab notebook. After each change, the notebook counts the result again (shape, totals, row counts) to show the change did what it should. Each sanity check has a comment saying what it found and what was decided.

Cleaning

Before changing anything, each table gets a look: size, types, duplicates, blanks, odd rows, labels and spaces. The aim is to see what's in the data before deciding what to fix.

Check	What I found	Action
Size and types (.shape, .info())	Country 12,776 × 8, product 10,848 × 10. Month is text. Year, value and weight are numbers.	One date built from month and year (step 3).
Duplicates (.duplicated())	None in either table.	None needed.
Blank weights (.isna())	110 in each table, all HMRC non-response estimates (EU exports 55, EU imports 55). The same rows in both tables: £37.2m, about 0.02% of value.	Kept and labelled (step 5).
'UK' as a country	1 row: £0, 0 kg. Its labels contradict each other: EU status, but a "Non EU - Exports" flow.	Dropped (step 4).
"Stores & Provis."	22 rows, £155,594 in total. Not a country.	Kept and labelled (step 5).
"Below Threshold Trade"	108 rows, £77.9m. No product detail: the same label sits in HS4, HS6 and CN8. EU flows only.	Kept and labelled (step 5).
Constant columns (.nunique())	In product, Commodities and HS2 hold one value each.	Dropped (step 6).
Country and product labels (.unique())	Country names mix full and short forms ("Cayman Islands", "Br Virgin Is"). The hs4 column is a code plus a very long description.	Country names left as HMRC wrote them; revisit if the dashboard uses a map. Product groups added (step 7).
EU and flow labels (.unique())	eu_status has 2 values and flow_type has 4, the same 4 in both tables. Each has one spelling, but non-EU is written two ways: "NON-EU" and "Non EU".	Left as HMRC wrote them.
Extra spaces (.str.strip())	None in any of the 12 text columns.	None needed.

Sanity checks

Six checks after cleaning, each with the reason it's there.

Check	Why	What I found	Decision
1. Negatives and extremes (.describe())	A negative or absurd value would throw off every total built on it.	No negative values. Zeros exist, and some weights are blank (see check 2). A few big rows pull the average well above the median.	Nothing to fix.
2. Zeros and blank weights	The value per kg can't divide by a blank or a zero, so I need to know which rows to leave out of it.	Value of 0: 14 country rows and 15 product rows (13 of each are estimates). Blank weight: the 110 estimated rows in each table. Weight of 0: 108 country rows and 75 product rows, none of them estimates.	Kept. Rows with money stay in the value totals. Value per kg leaves out blank and 0-weight rows.
3. Every flow has every month (.groupby(), .nunique())	A missing month would show up as a false dip in the monthly totals.	All four flows have all 55 months in both tables. Below-threshold rows have 54 in each EU flow: June 2026 is missing from both.	Nothing to fix. No June row added, because the data can't say what it would hold. No cause claimed.
4. EU and non-EU match the flow	Two columns say EU or non-EU. If they disagreed, the split would depend on which one I picked.	Each flow has one eu_status, and it matches the flow's name.	Nothing to fix.
5. Totals and duplicates (.sum(), .duplicated())	Cleaning shouldn't change any money, and a duplicated row would count the same trade twice.	Raw and cleaned totals are identical in both tables (£209,718,700,270). No duplicates.	Nothing to fix.

6. Spikes (biggest month ÷ median month)	One huge month can swing a trend, so each one needs a decision before the analysis uses it. See step 9.	Of 362 country and flow groups, 68 have a month above £10m, and two stand out. Israel, non-EU imports: eight months (seven in 2022, one in April 2023) between £78m and £278m, against a median month of £0.3m. The next largest is £2.5m. "Estimates", EU imports: HMRC's own estimate rows, with the largest months mostly in 2025 and 2026. The 294 groups at or under £10m weren't reviewed. They hold 2.8% of the money.	All rows kept. The data can't show whether these months are errors or real, so I make no claim about the cause. Both are flagged for Analyse.

Steps and code

The cleaning in order, with the reason for each step. The full code is in the project notebook.

Step 1: load the files and make working copies

Why: the original HMRC files stay untouched, so any figure can be checked back against them. All the cleaning happens on copies.

```
import pandas as pd

country = pd.read_csv('/content/drive/MyDrive/Project 3/HMRC Medicines by
Country (Jan 2022 - Jul 2026).csv')

product = pd.read_csv('/content/drive/MyDrive/Project 3/HMRC Medicines by
Product (Jan 2022 - Jul 2026).csv')

country_clean = country.copy()
product_clean = product.copy()
```

pd.read_csv reads a file into a table. .copy() makes a second, separate table, so nothing done to country_clean can touch country.

Step 2: tidy the column names

Why: HMRC's names have capitals, spaces and brackets, like "Value (£)" and "EU / NON EU". Short lowercase names are quicker to type and harder to get wrong.

```
country_clean = country_clean.rename(columns={
'EU / NON EU': 'eu_status',
'Continent': 'continent',
'Country': 'country',
'Value (£)': 'value_gbp',
'Net Mass (Kg)': 'net_mass_kg',
'Flow Type': 'flow_type',
'Year': 'year',
'Month': 'month'})
```

.rename(columns={...}) swaps each old name for a new one. The product table gets the same treatment.

Step 3: build one real date

Why: the question is about change over time. HMRC gives the month as a word and the year as a number, in two columns. Month names sort alphabetically (April comes before January), so they can't draw a timeline. A date can.

```
country_clean['date'] = pd.to_datetime(country_clean['month'] + ' ' +
country_clean['year'].astype(str), format = '%B %Y')
```

The first half joins the month and the year into text like "January 2022". .astype(str) turns the year into text first, because a number can't be joined to text. pd.to_datetime then reads the result as a date: %B is a full month name and %Y a four-digit year. The product table gets the same line. The result is 55 months, January 2022 to July 2026.

Step 4: drop the 'UK' row

Why: it's one row with £0 and 0 kg, and its labels contradict each other (EU status, but a "Non EU - Exports" flow). The UK has no place in a list of trading partners, and dropping a £0 row changes no total.

```
country_clean = country_clean[country_clean['country']!= 'UK']
```

The test inside the square brackets gives True or False for each row, and only the True rows stay. The table goes from 12,776 rows to 12,775.

Step 5: label the odd rows instead of deleting them

Why: four kinds of row aren't a country or a product: "Estimates" and "Stores & Provis." in the country table, "Non Response Estimates" and "Below Threshold Trade" in the product table.

They still hold real money. Deleting them would pull the totals away from HMRC's, and leaving them unlabelled would put "Estimates" in the list of countries. A row_type label lets each chart pick the rows it needs.

```
country_clean['row_type']='partner'

country_clean.loc[country_clean['country'] == 'Estimates', 'row_type'] = 'estimate'

country_clean.loc[country_clean['country'] == 'Stores & Provis.', 'row_type'] =
'stores'

product_clean['row_type'] = 'product'

product_clean.loc[product_clean['hs4'] == '30 🔸 Non Response Estimates',
'row_type'] = 'estimate'

product_clean.loc[product_clean['hs4'] == '30 HS2 Below Threshold Trade',
'row_type'] = 'below_threshold'
```

The first line adds a new column, row_type, and fills it with 'partner'. Each .loc line picks the rows that match and changes row_type in those rows only. The product table works the same way, starting from 'product'.

- **Country:** 12,643 partner, 110 estimate, 22 stores.
- **Product:** 10,630 product, 110 estimate, 108 below_threshold.

Step 6: drop the two constant columns

Why: Commodities and HS2 hold the same value in every row ("All Commodities" and "30 Pharmaceutical products"). A column that never changes tells you nothing about any row.

```
product_clean = product_clean.drop(columns = ['HS2', 'Commodities'])
```

.drop(columns = [...]) removes the columns in the list.

Step 7: name the product groups

Why: the hs4 column holds a four-digit product code followed by a very long description, far too long for a chart label. Six short names, one per code, keep the product charts readable.

```
group_names = {
'3001': 'Glands, organ extracts, heparin',
'3002': 'Blood products, antisera, vaccines',
'3003': 'Bulk mixed medicines',
'3004': 'Packaged medicines',
'3005': 'Dressings and bandages',
'3006': 'Other pharmaceutical products'}

product_clean['product_group'] = product_clean['hs4'].str[:4].map(group_names)
```

The first line is a dictionary: each code on the left, its plain name on the right. .str[:4] keeps the first four characters of each hs4 label, which is the code. .map(group_names) swaps each code for its name. The 218 estimate and below-threshold rows have no product detail, so they get no group.

Step 8: check that cleaning changed no money (check 5)

Why: cleaning should move and label rows, never change the money. If the cleaned totals differed from the originals, something had been dropped or doubled.

```
country_clean['value_gbp'].sum()
country['Value (£)'].sum()
```

.sum() adds up a column. Both give £209,718,700,270, and the same two sums on the product tables give the same figure.

Step 9: find the spikes (check 6)

Why: the project reads trends in exports and imports, and one huge month in one country can move a whole trend. Each unusual month needs a decision before the analysis builds on it. Comparing a country's biggest month with its own usual month finds them, whatever the country's size.

```
summary = country_clean.groupby(['flow_type',
'country'])['value_gbp'].agg(['median', 'max'])
summary['ratio'] = summary['max'] / summary['median']
summary[summary['max'] > 10000000].sort_values('ratio',
ascending=False).head(10)
```

The first line groups the rows by flow and country, then finds the median month and the biggest month for each group. The second divides the biggest by the median, so a ratio of 10 means the biggest month is ten times a normal one. The third keeps the groups whose biggest month is above 10,000,000 (£10m),

sorts them by ratio from largest to smallest and shows the first ten. Small groups can show a huge ratio just because their usual month is tiny, which is why the £10m cut-off is there. The 294 groups at or under it hold 2.8% of the money.

Step 10: save the cleaned tables
Why: Power BI reads the cleaned tables from files, and the originals stay as they are.

```
country_clean.to_csv('/content/drive/MyDrive/Project 3/country_clean.csv',
index=False)

product_clean.to_csv('/content/drive/MyDrive/Project 3/product_clean.csv',
index=False)
```

.to_csv(path, index=False) writes the table to a file. index=False leaves out pandas' row numbers, which would reach Power BI as an extra column.

Carried into Analyse

- Country breakdowns use partner rows only. "Estimates" and "Stores & Provis." aren't countries, so they never show up in a list of them.
- Value per kg leaves out blank and 0-weight rows, because there's no weight to divide by.
- The imports trend and the gap are tested with and without Israel's eight spike months, to see how much of the answer they drive.
- June 2026 has no below-threshold rows in either EU flow. It's on record, with no cause claimed.

Output

Two cleaned files for the Power BI dashboard, saved in the Project 3 folder: country_clean.csv (12,775 rows, 10 columns) and product_clean.csv (10,848 rows, 11 columns). The original HMRC files are untouched.

4. ANALYSE PHASE

Approach

The analysis follows the minister's three questions. Each gets one main measure, set in the Ask phase.

Question	Measure	Why
Selling more or fewer medicines abroad?	Monthly export value, a 12-month average, and the change in that average from the same month a year earlier.	Single months swing too much to read by eye. The 12-month average smooths them out. Each 12-month average covers every month of the year once, so two averages a year apart are like for like.
Who is buying?	Each country's share of exports, year by year.	A share shows who matters, whatever the overall level. Comparing years shows whether it has changed.
Leaning more on imports?	The monthly gap (exports minus imports), averaged over 12 months as in the first row, and each country's share of imports.	Above zero, the UK sells more than it buys. Below zero, the reverse. If the gap moves further below zero, the UK leans more on imports. The shares show who supplies them.
Supporting check: price or volume?	Value per kg.	Value alone can't say whether a change is more goods or higher prices. Weight gives a rough check.

Totals use every row, so they match HMRC's. Country charts use partner rows only, and product charts use product rows only. Value per kg leaves out blank and 0-weight rows. Each rule comes from a decision in the Process phase.

Steps and code

The calculations in order, with the reason for each step. They start from the two cleaned tables, country_clean and product_clean. This is descriptive analysis: it shows what moved, not why. The full code is in the project notebook.

- Steps 1 and 2: exports, imports and the gap by month, then smoothed (questions 1 and 3).
- Steps 3 and 4: the countries that buy and the countries that supply (questions 2 and 3).
- Step 5: a test of whether eight unusual months drive the result.
- Steps 6 and 7: a look behind the totals, by medicine and by price or volume.
- Step 8: the tables saved for the Share phase.

Step 1: build one row per month

Why: questions 1 and 3 compare exports, imports and the gap over time, so each needs one value per month.

```
monthly = country_clean.pivot_table(values='value_gbp',
columns='flow_type', aggfunc='sum', index='date')
monthly['exports'] = monthly['EU - Exports'] + monthly['Non EU - Exports']
monthly['imports'] = monthly['EU - Imports'] + monthly['Non EU - Imports']
monthly['gap'] = monthly['exports'] - monthly['imports']
```

pivot_table works like a pivot table in Sheets: one row per date, one column per flow, with value_gbp added up in each cell. The next three lines add the EU and non-EU parts together, then take imports from exports to get the gap. Above zero, the UK sold more than it bought.

The result has 55 rows, one per month. Its four flow columns add up to £209,718,700,270, the same as country_clean, so no money was lost or counted twice.

Step 2: average over 12 months and compare with the year before

Why: a single month is too noisy to show a trend. A 12-month average covers every month of the year once, so two averages a year apart are like for like.

```
monthly['exports_12m'] = monthly['exports'].rolling(12).mean()
monthly['imports_12m'] = monthly['imports'].rolling(12).mean()
monthly['gap_12m'] = monthly['gap'].rolling(12).mean()
monthly['exports_yoy'] = monthly['exports_12m'].pct_change(12)
monthly['imports_yoy'] = monthly['imports_12m'].pct_change(12)
monthly['gap_yoy_gbp'] = monthly['gap_12m'].diff(12)
```

.rolling(12).mean() averages each month with the 11 before it, like an AVERAGE over a sliding range of 12 cells in Sheets. .pct_change(12) compares each row with the one 12 rows above, which is the same month a year before, so it gives the year-on-year change (yoy) as a fraction (-0.02 means -2%). .diff(12) subtracts the value 12 rows above, which gives the change in pounds. The gap uses pounds because it changes sign, so a percentage would be meaningless.

The first 11 months are blank (NaN) in the 12-month columns, because they do not have 12 months behind them. The first 23 are blank in the year-on-year columns, because each change needs two 12-month averages, 12 months apart. For exports, imports and the gap, the first 12-month average matches a plain average of the same 12 months.

Step 3: who buys the exports

Why: question 2 asks which countries buy the exports and whether that has changed. A share shows who matters whatever the overall level, and one column per year shows whether it moved.

```
export_partners = country_clean[(country_clean['row_type'] == 'partner') &
(country_clean['flow_type'].isin(['EU - Exports', 'Non EU - Exports']))]
export_by_country =
export_partners.groupby('country')['value_gbp'].sum().sort_values(ascending
=False)
export_share = export_by_country / export_by_country.sum()
export_by_country_year = export_partners.pivot_table(index='country',
columns='year', values='value_gbp', aggfunc='sum')
export_share_by_year = export_by_country_year /
export_by_country_year.sum()
```

Only partner rows are used, because estimates and stores are not countries. Partner rows hold 99.98% of export value. In the first line, & keeps only the rows that pass both tests, and .isin([...]) is True for the rows whose flow is in the list. .groupby('country') adds up the rows of each country, and .sort_values(ascending=False) puts the biggest first. Dividing by the total gives each share as a fraction (0.25 means 25%).

Dividing the by-year table by its own .sum() divides each year by the total for that year, so every year adds up to 1, including 2026, which holds January to July only.

The 2023 swing runs through the year. Month by month, the United States share is above its 2022 share (22.7%) in 11 of the 12 months of 2023, and Germany's is below its 2022 share (13.1%) in 11 of 12.

Step 4: who supplies the imports

Why: question 3 also asks who supplies the imports. The method is the same as Step 3, with the two import flows.

```
import_partners = country_clean[(country_clean['row_type'] == 'partner') &
(country_clean['flow_type'].isin(['EU - Imports', 'Non EU - Imports']))]
import_by_country =
import_partners.groupby('country')['value_gbp'].sum().sort_values(ascending=False)
import_share = import_by_country / import_by_country.sum()
import_by_country_year = import_partners.pivot_table(index='country',
columns='year', values='value_gbp', aggfunc='sum')
import_share_by_year = import_by_country_year / import_by_country_year.sum()
```

Partner rows hold 99.98% of import value. HMRC records each import under its country of dispatch, the country it was exported from to come to the UK. That may not be where it was made, so the shares show who dispatched the imports, not who made them.

The 2025 swing also runs through the year. Belgium's share is between 7.3% and 12.3% in every month of 2025. France is above its 2024 share in all 12 months and Spain in 11 of 12, with Spain peaking in May. The United States share is below its 2024 share in 11 of 12 months, and has been mostly low since August 2024.

Step 5: test whether eight unusual months drive the result

Why: check 6 found eight months when imports from Israel were far above its usual month. Dropping them from a copy shows how much of the answer they drive, and the main tables stay whole.

```
israel_spikes = country_clean[(country_clean['country'] == 'Israel') &
(country_clean['value_gbp'] >= 10000000) & (country_clean['flow_type'] ==
'Non EU - Imports')]
country_no_israel_spikes = country_clean.drop(israel_spikes.index)
monthly_no_israel_spikes =
country_no_israel_spikes.pivot_table(values='value_gbp',
columns='flow_type', aggfunc='sum', index='date')
```

israel_spikes picks the Israel rows in Non EU - Imports worth £10m or more. There are 8, worth £1,023,513,911 in all. .drop(israel_spikes.index) removes those rows from a copy by their row labels, and a check confirms the money removed is exactly the money in those 8 rows. Steps 1 and 2 are then repeated on the copy for imports and the gap, and Step 4 for the supplier shares.

The other group flagged in check 6, the Estimates rows in EU imports, holds £18.6m in all (under 0.01% of value). Taking it out would change no 12-month average by more than £1.6m, so it stays in.

Step 6: split the totals by medicine

Why: a change in the total can come from one kind of medicine. The six product groups from the Process phase show which.

```
product_rows = product_clean[product_clean['row_type'] == 'product']
product_exports = product_rows[product_rows['flow_type'].isin(['EU -
Exports', 'Non EU - Exports'])]
product_imports = product_rows[product_rows['flow_type'].isin(['EU -
Imports', 'Non EU - Imports'])]
```

```
export_by_group_year = product_exports.pivot_table(index='product_group',
columns='year', values='value_gbp', aggfunc='sum')
import_by_group_year = product_imports.pivot_table(index='product_group',
columns='year', values='value_gbp', aggfunc='sum')
gap_by_group_year = export_by_group_year - import_by_group_year
```

Estimate and below-threshold rows belong to no group, so only product rows are used. Product rows hold all but £115m (0.05%) of the value. The two pivot_table lines work as in Step 3, with product groups down the side. Subtracting one table from another lines up the rows and columns by their labels, so each group and year gets its own gap, and a check confirms the gap table adds up to total exports minus total imports. A further table, group_totals, adds up each group over the whole period and gives each group a share of exports and of imports.

Bulk mixed medicines imports from outside the EU were also ranked by month: the 8 biggest months are the 8 Israel months. June 2026 has no below-threshold rows in either EU flow, but these tables leave those rows out in every month, so it does not change them.

Step 7: work out value per kg and what moved it

Why: value alone cannot say whether a change is more goods or higher prices. Value per kg is a rough check, and splitting the change in value into kilos and value per kg shows which one moved.

```
country_with_weight = country_clean[country_clean['net_mass_kg'] > 0]
value_by_month['exports_per_kg'] = (value_by_month['exports_12m']) /
(kg_by_month['exports_12m'])
value_by_month['exports_value_yoy'] =
value_by_month['exports_12m'].pct_change(12)
value_by_month['exports_kg_yoy'] =
kg_by_month['exports_12m'].pct_change(12)
value_by_month['exports_per_kg_yoy'] =
value_by_month['exports_per_kg'].pct_change(12)
```

Value per kg needs a weight, so country_with_weight keeps the rows with a weight above zero. The rows left out hold £41,061,896 (0.02% of value) and stay in every value table. value_by_month and kg_by_month are built from these rows as monthly was in Step 1, with 12-month totals (.rolling(12).sum()) in place of averages. Value per kg is the 12-month value divided by the 12-month kilos.

The three year-on-year changes are linked: 1 plus the change in value equals 1 plus the change in kilos, times 1 plus the change in value per kg. So the split shows which one moved. Imports follow the same lines, and are worked out again without the 8 Israel rows, which carry a lot of money for very little

weight. Value per kg also moves when the mix of medicines changes, so it is named value per kg, not price.

Step 8: save the tables for the Share phase

Why: the Share phase reads these files, so the charts use exactly the numbers worked out here.

```
monthly.to_csv('/content/drive/MyDrive/Project
3/analysis_tables/monthly.csv')
export_share_by_year.to_csv('/content/drive/MyDrive/Project
3/analysis_tables/export_share_by_year.csv')
value_by_month.to_csv('/content/drive/MyDrive/Project
3/analysis_tables/value_by_month.csv')
```

.to_csv(path) writes a table to a file. The same line, with a different name, saves each of the 13 tables. Unlike the cleaned tables in the Process phase, the index is kept, because the dates, countries and product groups are held in the index.

Summary of the analysis

The UK is selling less abroad by value than in 2022. Since a near balance in 2023 it has bought more than it has sold, by a wider margin each year. Monthly amounts are averages over the 12 months to the date shown. Yearly amounts are calendar-year totals.

Question or check	What the data shows	Analyse steps
Selling more or fewer medicines abroad?	Less in value. Exports averaged £1,884m a month in 2022 and £1,749m in 2025, lower at every December than the one before. The 12 months to Jul 2026 average £1,703m, 7.9% below the 12 months to Jul 2025.	2
Who is buying?	The United States, Belgium and Germany, with 25.2%, 11.6% and 10.4% of export value over the whole period. The 5 biggest buyers over the whole period hold 59% to 63% in every year and the 10 biggest 75% to 79%. The United States rose from 22.7% in 2022 to 29.2% in 2023 and fell back to 23.3% in 2024, while Germany fell from 13.1% to 7.2% and rose again to 9.4% in 2024. Germany reaches 10.7% in 2025 and 12.9% in Jan–Jul 2026 (part year).	3

Leaning more on imports?	Imports fell 17.5% in 2023, from £2,223m a month to £1,834m, then rose to £1,995m in 2025. In the 12 months to Jul 2026 they are £1,976m, down 0.8% on the 12 months to Jul 2025. The gap was -£339m a month at Dec 2022, +£10m at Dec 2023 and -£273m at Jul 2026. It reached balance in 2023, and each December since has been wider than the one before. The Dec 2022 figure includes seven of the eight Israel months. Without those months it is -£261m, close to the Jul 2026 figure.	2 and 5
Who supplies the imports?	Belgium, the United States, the Netherlands and Germany, with 13.7%, 12.9%, 12.6% and 12.0% of import value over the whole period. The 5 biggest suppliers over the whole period hold 50% to 64% by year and the 10 biggest 84% to 86%. In 2025 the share from Belgium fell from 16.4% to 9.7% and from the United States from 15.5% to 9.3%, while the share from France rose from 5.2% to 9.1% and from Spain from 2.7% to 7.0%. In Jan–Jul 2026 France is back at 5.3% and Spain at 2.1% (part year). Ireland rose from 6.9% in 2024 to 8.7% in 2025 and is at 15.0% in Jan–Jul 2026 (part year).	4
Which medicines?	Over the whole period, packaged medicines are 69.5% of export value and 61.8% of import value, and blood products, antisera and vaccines 25.6% and 32.3%. Exports of packaged medicines fall every year, from £16.8bn in 2022 to £13.9bn in 2025, while exports of blood products rise from £4.8bn to £5.9bn. The balance on packaged medicines swings from a £631m surplus in 2022 to a £1,633m deficit in 2025, while the deficit on blood products narrows from £3,594m to £1,234m.	6
Price or volume?	In the latest 12 months exports fell 7.9% in value. The fall is in value per kg (down 9.5%), not in weight (kilos up 1.8%). In 2023 exports fell 2.1% in value: kilos fell 8.1%, partly offset by a 6.6% rise in value per kg. Imports fell 17.5% in 2023: kilos fell 10.2% and value per kg fell 8.1% (5.2% without the Israel months). Value per kg also moves with the mix of medicines, so it is not a price.	7
Do the eight Israel months drive the answer?	They change the size of the 2023 figures, not the direction. The eight months are worth £1.02bn. Without them the 2023 fall in imports is 14.9% instead of 17.5%, and the 2023 rise in the gap is £278m instead of £349m. 2024 differs only a little and 2025 onwards is identical. The supplier shares move only in 2022 and 2023. In 2022 Israel falls from 3.5% to 0.02% of import value. The large 2022 figure for bulk mixed medicines imports (£1,063m, against £303m in 2023) comes from the same months.	5 and 6

Customs data shows what moved, not why. It cannot show what medicines cost the NHS or what caused any change.

Carried into Share

- Calendar years come from the December rows. 2026 holds January to July only, so show the latest 12 months from the Jul 2026 row and label 2026 as a part year, shares included. A January to July share can differ from a full-year share.

- Shares are fractions, so 0.25 means 25%. An empty cell in a country-by-year table means no trade was recorded that year.

- Quote the figure without the eight Israel months beside the 2023 fall in imports and the 2023 rise in the gap. The test covers total imports, the gap, the supplier shares and import value per kg, not the non-EU import line on its own.

- Name value per kg as value per kg, not price, and report the latest fall in exports as a fall in value per kg, not in weight.

- Import countries are countries of dispatch, not makers. Label them that way.

- Report the 2023 swing in export shares and the 2025 swing in import shares as recorded, with no cause. They are yearly figures, and each runs through the year, not one month.

- Amounts are not adjusted for inflation.

Output

The 13 tables are saved as CSV files in the analysis_tables folder inside the Project 3 folder, for the Power BI dashboard: monthly totals, buyer and supplier values and shares by country and year, product group totals by year and flow, and value per kg. Three of them hold the same figures without the eight Israel months.

5. SHARE PHASE

Audience & Formats

The audience is the minister and Treasury officials, who are not data specialists. The findings come in three formats, so each reader can go as deep as they need: a one-page executive brief, a slide deck and an interactive Power BI dashboard.

The dashboard

Contrast comes from colour: blue is exports, orange is imports and grey is context, and blue and orange stay clearly apart for colour-blind readers. Each chart has a title that states its finding and a subtitle that says what is plotted, and columns start at zero. Page 2 shows the five biggest buyers and four biggest suppliers over the period, plus France and Spain (the two biggest rises in 2025) and Ireland (the highest share in Jan–Jul 2026).

Key Findings

The findings answer the minister's three questions.

Exports are worth less than in 2022. They averaged £1,884m a month in 2022 and £1,703m in the 12 months to Jul 2026.

Imports now outweigh exports. The gap was a near balance in 2023 and has widened every year since, to -£273m a month in the 12 months to Jul 2026.

The biggest buyers and suppliers swung in 2023 and 2025. In 2023 the US share of exports rose from 22.7% to 29.2% and Germany's fell from 13.1% to 7.2%. In 2025 the import shares of Belgium and the US each fell by over 6 points.

The latest fall in export value is in value per kg, not kilos. In the 12 months to Jul 2026, value fell 7.9% and kilos rose 1.8%, so value per kg fell 9.5%.

Reading the Results

Customs data shows what moved, not why, so no finding gives a cause. Import countries are countries of dispatch, not makers. 2026 covers January to July only. Eight unusually large import months from Israel (£1.02bn) change the size of the 2022 and 2023 figures, not their direction. Amounts are not adjusted for inflation.

6. ACT PHASE

Conclusion

UK medicine exports are worth less than in 2022. Since a near balance in 2023 the UK has bought more than it has sold, by a wider margin each year. Packaged medicines, the biggest group, fell £2.9bn from 2022 to 2025 while blood products rose £1.1bn, and the latest fall in export value is in value per kg, not kilos.

Recommendations

The three items say where to look next, not what to decide.

Where to look	Why	Next step
Packaged medicines	The biggest group (69.5% of export value over the whole period). Its exports fell every year, from £16.8bn in 2022 to £13.9bn in 2025.	Split them by product and by buyer.
Value per kg	In the latest 12 months export value fell 7.9% while kilos rose 1.8%, so value per kg fell 9.5%.	Compare each product's export value against its weight sold, to identify which products have the highest and lowest value per kg.
Main buyers and suppliers	Over the whole period the top three buyers took 47% of exports, and each of the top four suppliers sent 12–14% of imports.	Track each country's share every quarter.

What success looks like: the department can say which products and buyers explain the fall in export value, and why value per kg fell.

The analysis informs the briefing. It does not decide the spending.

Next Steps & Further Exploration

More deliverables: a product-level view of packaged medicines by buyer, a table of value per kg by product, and a quarterly tracker of buyer and supplier shares.

More data: prices by product, to tell price from product mix, and where imported medicines are made.